



Targeting Alpha-Synuclein in Parkinson's Disease

**Do small-molecule inhibitors of protein aggregation
also prevent its accumulation with lipids?**

Keywords: alpha-synuclein, protein aggregation, Parkinson's Disease, lipids, cellular assay





Our Abstract:

Parkinson's Disease (PD), a neurodegenerative disorder, is characterized by the pathological aggregation of α -synuclein (α S) into Lewy bodies, primarily composed of fibrillar and lipid/membrane-rich α S inclusions. Recent studies suggest that both fibrillar and lipid-rich α S aggregates may contribute to PD pathology. This study investigates how lipid-rich α S inclusions are affected by small molecules that had previously been shown to disaggregate fibrillar α S protein aggregation. Using a live-cellular assay based on YFP-tagged membrane-accumulating, engineered α S-3K (E35K+E46K+E61K) in M17D neuroblastoma cells, we assessed the effects of fasudil, Epigallocatechin gallate (EGCG), anle138b, and ethanol compared to the positive control trifluoperazine (TFP), which is known to disassemble lipid-rich α S aggregates. fasudil, EGCG, and anle138b had previously been shown to prevent protein aggregation of α S. In our experiments, fasudil and anle138b significantly decreased inclusion formation in a dose-dependent manner in both α S lipid-rich inclusion prevention and reduction assays, while EGCG and ethanol exhibited either negligible or inclusion-increasing effects. These findings indicate that fasudil and anle138b may effectively target lipid-rich α S inclusions, supporting further exploration of their therapeutic potential in PD. However, fasudil's observed associated toxicity necessitates careful evaluation. This study underscores the therapeutic relevance of targeting diverse α S aggregate types in PD.



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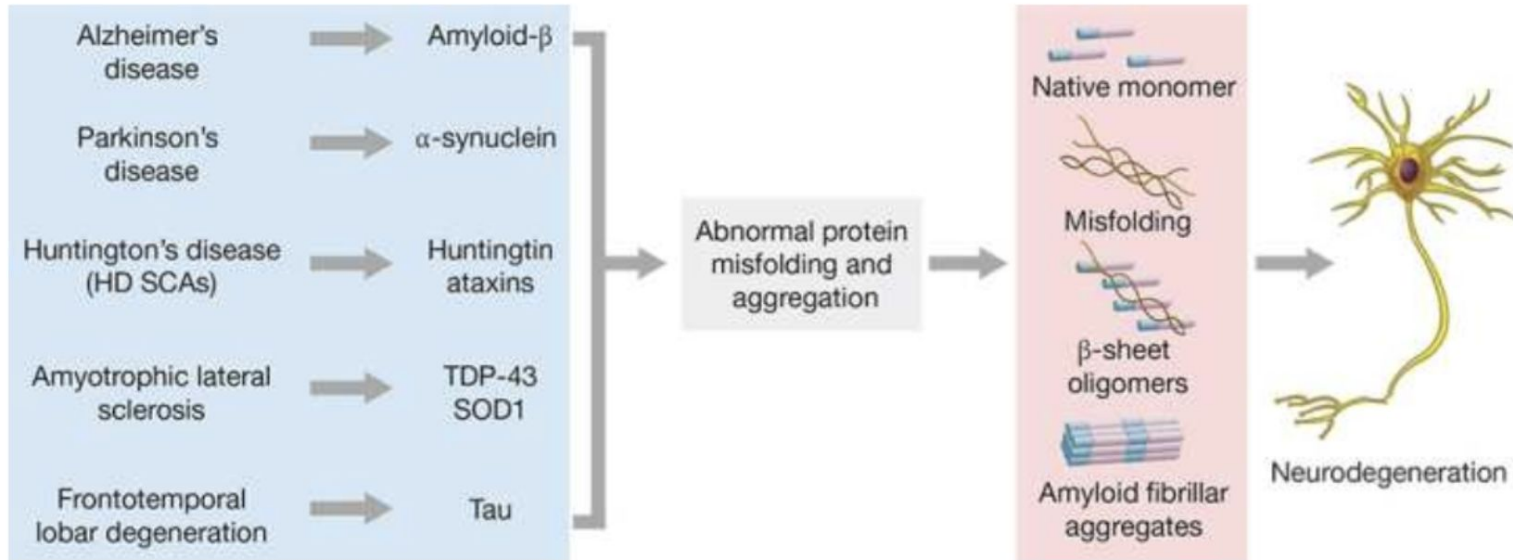
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01

Introduction

Protein Misfolding Diseases



Parkinson's Disease

A neurodegenerative disorder affecting the nervous system

Key symptoms include:

1. Stiffness
2. Slowing of movement
3. Tremors (often at rest)

May also cause:

1. Balance problems
2. Speech & writing changes

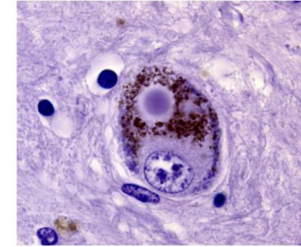
Control



PD



Loss of Dopaminergic
Neurons in the Substantia
Nigra (Midbrain)



'Lewy Body':
aggregated **α -synuclein (α S)**

Alpha-Synuclein: Lipid-rich or Fibrillar?

Early Research:

Lewy bodies primarily composed of β -sheet structured fibrillar α -synuclein (α S) (Ref: 9)

Recent Findings:

Lewy bodies also contain lipid/membrane-rich α S aggregates

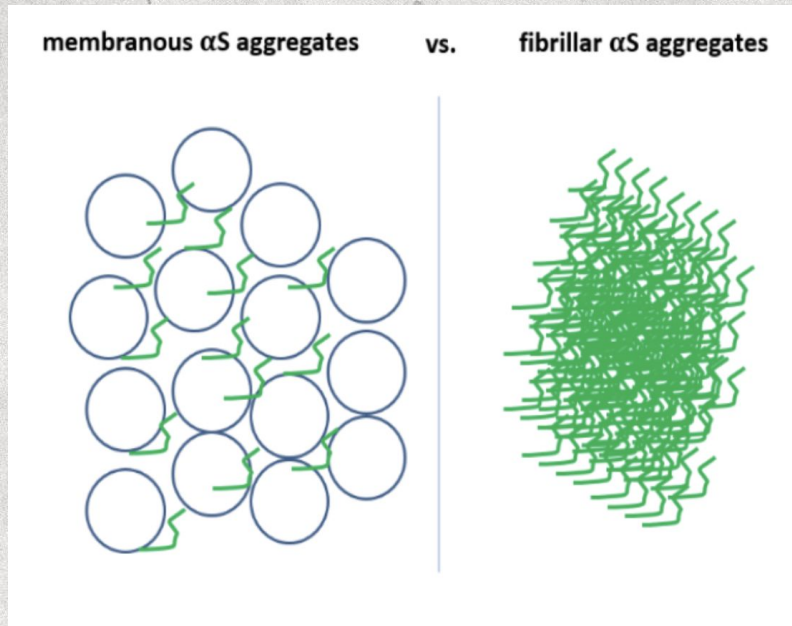
Fibrillar α S is not the dominant species in these aggregates

Other folding states of α S may be more prevalent

Current Debate:

Relative contribution of fibrillar vs. lipid-rich α S aggregates to pathology remains unclear

Evidence suggests both types may play a role in disease progression



(Fanning et al., 2020)





Our Study: Purpose

Current PD Treatment Gap:

No existing treatments target Parkinson's disease (PD) pathology directly.

Most research focuses on fibrillar α -synuclein (α S) aggregates, but lipid-rich α S aggregates remain understudied.

Therapeutic Potential of Small Molecules:

Some compounds (e.g., Fasudil, EGCG, anle138B) can disrupt fibrillar α S aggregates

Unknown: Do they also work on lipid-rich α S aggregates?

Study's Unique Approach:

First to test anti-aggregation compounds on lipid-rich α S inclusions using a live-cell assay (α S-3K:: YFP).

Compares effects of: Fasudil, EGCG, anle138B, Trifluoperazine (positive control), DMSO, Ethanol

02

Materials & Methods



Cell Culture:

Cell Line: M17D cells

Growth Medium: Dulbecco's Modified Eagle's Medium (DMEM)

Supplemented with: 10% Fetal Bovine Serum (FBS)

Antibiotics:

- 50 units/mL penicillin
- 50 μ g/mL streptomycin
- 2 mM L-glutamine

Induction of α S-YFP Expression: Treated with 1 μ g/mL doxycycline



Compounds:

Positive Control:

Trifluoperazine (TFP)–Sigma-Aldrich (Catalog #: T8516, 2019)

Test Compounds:

Epigallocatechin gallate (EGCG)–Fisher Scientific (Catalog #: 50-163-8008)

Fasudil–Fisher Scientific (Catalog #: 50-187-2031)

Anle138b–Fisher Scientific (Catalog #: 50-196-7890)

Solvent Control:

Ethanol (EtOH)–Sigma-Aldrich (Catalog #: 493546-500ML)



Drug Treatment:

Prevention Assay

Day 1: Plate cells

Day 2: Treat with test compounds

Day 3: Induce α S-3K::YFP expression

Day 5: Live-cell microscopy analysis

Clearance Assay

Day 1: Plate cells

Day 2: Induce α S-3K::YFP expression

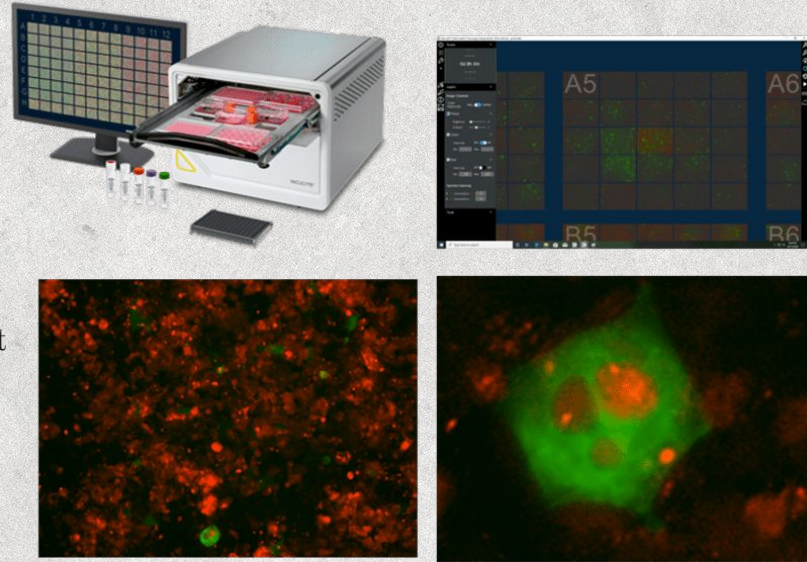
Day 4: Treat with test compounds (after confirming inclusion populations)

Days 5-6: Live-cell microscopy analysis

Imaging (Incucyte)

Imaging (Incucyte). Inclusion formation was recorded at 4-hour intervals in α S-3K::YFP cells on a 96-well plate using an IncuCyte S3-C2 machine (Essen Bioscience). The following parameters were used in the processing definition “Inclusions6”.

Segmentation: Surface Fit; Threshold (GCU): 2.0000; Edge Split On; Edge Sensitivity: 100; Hole Fill (μm^2): 0.0000; Adjust Pixel Size: -2; Area (μm^2) - min: (none), max: 40.000; Eccentricity - min: (none), max: 0.8500; Mean Intensity - min: 120.00, max: (none); Integrated Intensity: (none).



03

Results

Two Assays



Prevention Assay

After plating (5,000 cells/well), a 24-hr drug treatment (0-30 μ M) and 24-hr doxycycline induction preceded inclusion imaging at 48 hr, with data normalized to DMSO.

After initial screening, compounds were evaluated for clearance of pre-formed inclusions by treating cells 24 hr post-induction of α S-3K:: YFP expression.

Reduction Assay



Statistical Analysis:

n = 36 replicates per condition

Significance vs. DMSO (0 μ M):

****P < 0.0001, ***P < 0.001, **P < 0.001 (Kruskal-Wallis test)

n.s. = not significant



Prevention: Representative Treatment Images (YFP Signal)

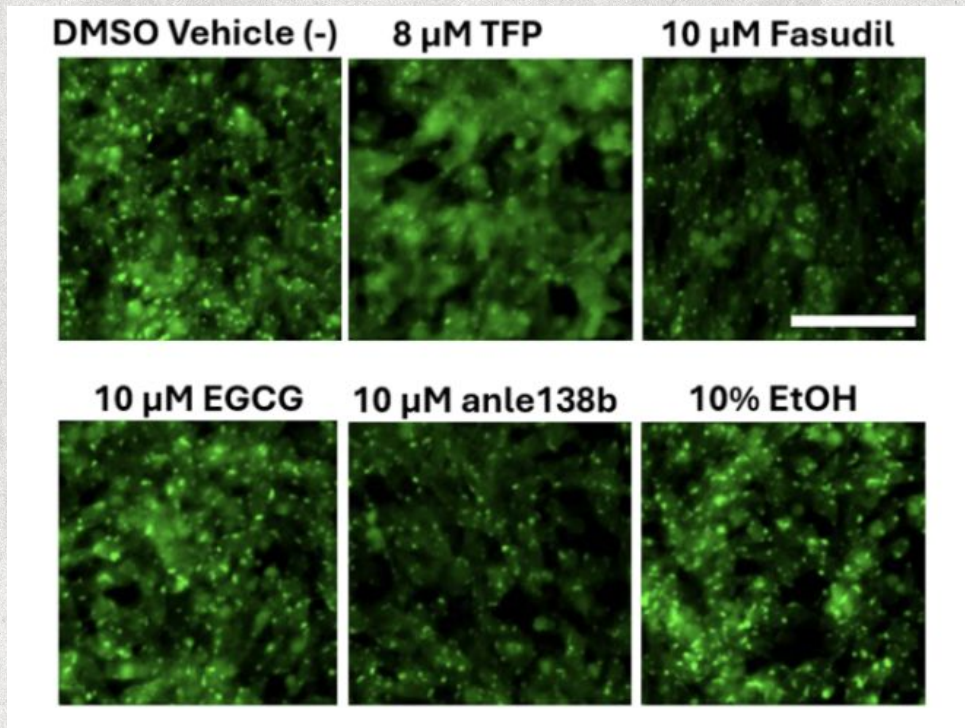
Conditions Shown:

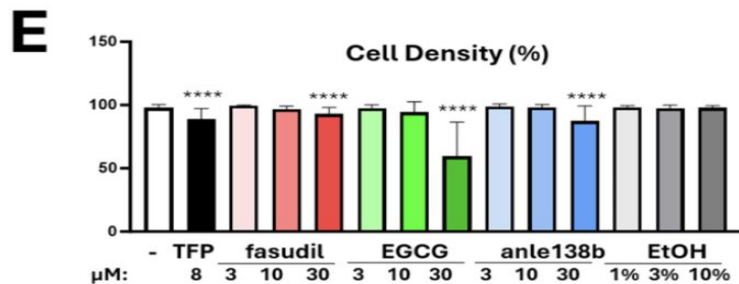
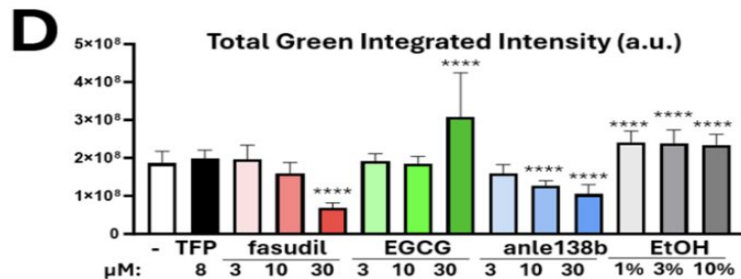
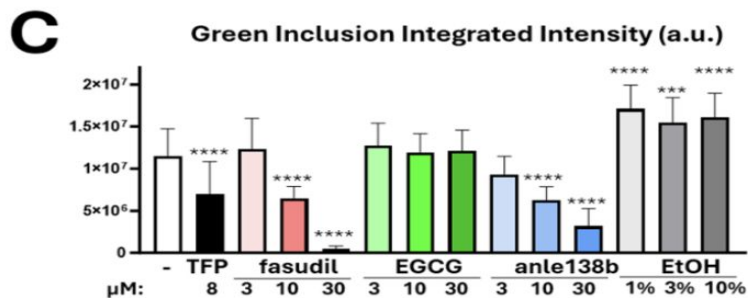
- DMSO (vehicle control)
- 8 μ M TFP (positive control)
- 10 μ M Fasudil
- 10 μ M anle138b
- 10 μ M EGCG
- 10% EtOH

Induction: α S-3K:: YFP expression triggered (doxycycline)

Imaging: Performed 24h post-induction

Scale Bar: 100 μ m





Quantification of Drug Effects on α S-3K::YFP Inclusions

C) Dose-Response Analysis Tested Compounds & Concentrations:

- TFP (positive control): 8 μ M
- Fasudil: 3, 10, 30 μ M
- EGCG: 3, 10, 30 μ M
- Anle138b: 3, 10, 30 μ M
- EtOH: 1%, 3%, 10%

*D) Total Green Signal Quantification
Measured YFP intensity across same treatments*

*E) Cell Density Assessment
Verified compound toxicity/viability in parallel*



Reduction: Representative Treatment Images (YFP Signal)

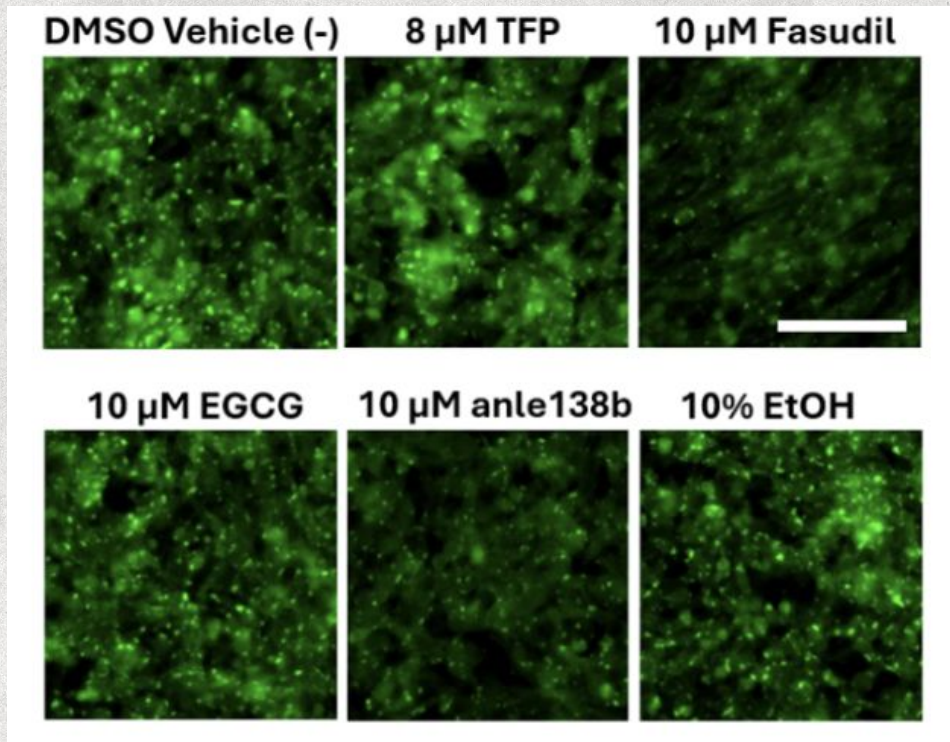
Conditions Shown:

- DMSO (vehicle control)
- 8 μ M TFP (positive control)
- 10 μ M Fasudil
- 10 μ M anle138b
- 10 μ M EGCG
- 10% EtOH

Treatment: Drugs added 24 hours post-induction

Imaging: Performed 24 hours post-treatment

Scale Bar: 100 μ m





Reduction Assay

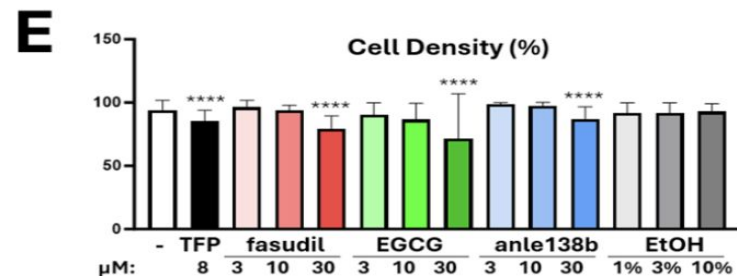
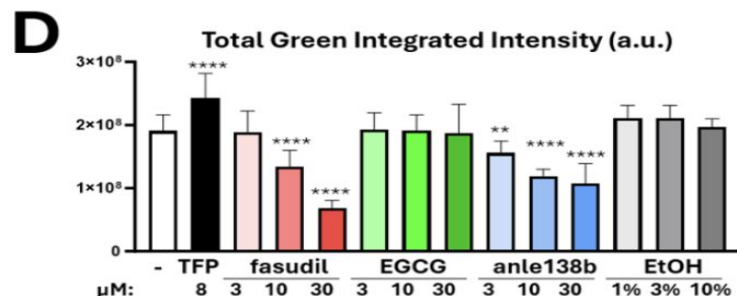
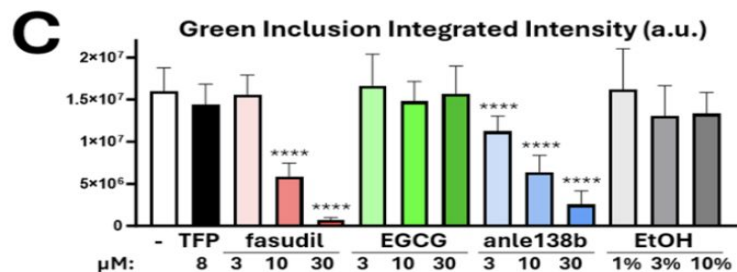
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D) Total Green Signal Quantification Measured YFP intensity across same treatments

E) Cell Density Assessment Verified compound toxicity/viability in parallel



04

Discussion



Discussion

1

Promising Drugs

Trifluoperazine (TFP), anle138b, and Fasudil reduced inclusion intensity.

Ethanol and EGCG showed no significant effects.

TFP:

- Dissolves existing inclusions but does not reduce total α S levels.

Fasudil/anle138b:

- Target both aggregation and degradation pathways.

2

Notable Performance of Fasudil & anle138b

Previously known to disaggregate fibrillar α S, now effective against lipid-rich α S aggregates.

Fasudil:

Rapidly reduces both inclusions and total α S levels (within 4 hours).

anle138b:

First dissolves inclusions (4h treatment), then may promote degradation.

Delayed reduction in total α S levels.



Discussion

3

Validation

No YFP signal quenching:

Lack of immediate green signal reduction confirms drug effects are not fluorescence artifacts.

Dose optimization:

30 μ M: Toxic for all drugs (reduced cell density).

10 μ M: Effective for both Fasudil & anle138b (reduced inclusions & total α S).

3 μ M: Only anle138b showed efficacy.

4

Therapeutic Potential & BBB Penetration

anle138b:

Blood-brain barrier (BBB) permeable (prior evidence).

Effective at low doses (3 μ M).

Fasudil:

BBB-protective properties and enhances cell impedance.

Requires higher doses (≥ 10 μ M) for efficacy.



Discussion

5

Clinical Considerations

Toxicity:

Both drugs need further safety studies for doses $>10\text{ }\mu\text{M}$.

IC₅₀ estimate:

5–15 μM likely dissolves $\geq 50\%$ inclusions (based on trend).

6

Implications

Dual targeting:

Both drugs disrupt β -sheet fibrils and lipid-rich aggregates.

May also trigger αS degradation.

Hypothesis:

Lipid-rich aggregates might contain hidden fibrillar structures, explaining drug efficacy.



Discussion

7

Suggested Methods for Future Work

Sequential protein extraction + Western blot to quantify:

Membrane-associated vs. cytosolic α S.

Changes in α S60:14 (multimer:monomer) ratios.

9

Clinical Relevance

Understanding α S redistribution could:

Reveal drug-specific mechanisms.

Identify toxicity risks from monomer buildup.

8

Key Hypotheses from Current Data

Reduced inclusions likely decrease membrane-bound α S (based on past studies).

Fasudil/anle138b may:

Preferentially degrade one pool (cytosol vs. membrane).

Alter toxic monomer accumulation.

05

Conclusion

Conclusions

Objective

Tested compounds known to disaggregate fibrillar α S for efficacy against lipid-rich α S inclusions.

Used prevention and reduction assays in α S-3K:: YFP neuroblastoma cells.

Implications for PD Therapy

Pathology-independent therapeutic strategies.

Dual-mechanism candidates for PD drug development.



Key Findings

Effective Compounds:

Fasudil & anle138b:

Reduced inclusion intensity dose-dependently in both assays.

Showed potential for broad-spectrum α S disaggregation.
Ineffective Compounds:

EGCG: No change in inclusions.

Ethanol: Increased inclusion formation.



06

Acknowledgement & References



Thanks

I would like to personally thank Dr. Ulf Dettmer and his lab for allowing me to not only visit in person to conduct my experiment in his facilities but also serve as my mentor throughout the duration of this research project.



Journal Link

Publication in the Journal of High School Science

<https://jhss.scholasticahq.com/article/131827-parkinson-s-disease-linked-protein-synuclein-do-small-molecule-inhibitors-of-protein-aggregation-also-prevent-its-accumulation-with-lipids>

Testimonial – Upper Canada College

<https://jhss.scholasticahq.com/pages/776-metrics>



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